

Mine Valuation

B S Choudhary

Department of Mining Engineering

IIT(ISM) Dhanbad



Contents

- ▶ Basic concept,
- ▶ Earlier approaches to mine valuation,
- ▶ Recent approaches to valuation;
- ▶ Time value of money,
- ▶ Price information - revenue estimates, annuity, Accounting profits and cash flows.



Basic concept

- ▶ The term mine valuation implies the assigning of a Rupees or other currency value to the worth of a mine or mining project and provides a measure of the desirability of ownership of that property.
- ▶ Value is the amount of money which a purchaser desires to pay for the ownership of a mineral property. The value arrived at is the mine value, i.e., today's money of future earnings that will be generated by the mine/mineral property worth.
- ▶ The valuation of a mineral property involves gathering number of information's, like geology, mining, market, etc. valuation differs from evaluation, where evaluation simply focuses on the technical aspects of an asset, and valuation focuses on value of the asset.

As such, several types of value may be encountered in performing a mine valuation study. These are

- ▶ 1. Market value.
- ▶ 2. Full cash value.
- ▶ 3. Salvage value.
- ▶ 4. Replacement value.
- ▶ 5. Capitalized value.
- ▶ 6. Book value.
- ▶ 7. Assessed value.
- ▶ 8. Insured value.

Salvage Value: Salvage is the net sum, over and above the Cost of removal and sale, realized for a property or asset when it is retired from service. Salvage value and scrap value are synonymous when the property or asset retired from service is scrapped for the value of its materials.

Replacement Value: Replacement value refers to the existing value of a property or asset as determined on the basis of what it would cost to replace the property or its service with at least equally satisfactory and comparable property and service. The concept of replacement value is fundamental to the cost approach utilized by appraisers.

Book Value: Book value is the original investment in the property or asset as carried on the organization's books less any cumulative allowance for depreciation or amortization entered on the books.

Assessed Value: The assessed value of a property is the value entered on the official assessor's records as the value of the property applicable in determining the amount of ad valorem taxes to be paid by the property owner.

Insured Value: The insured value of a property refers to that value at which the property has been insured against loss or disaster. This value is generally associated with replacement value for tangible assets and earning capacity for property such as mines (ore deposits).

Capitalized Value: The capitalized value of a property is the sum of discounted future annual net earnings generated by the property. The capitalized value concept is synonymous with the income approach to value estimation for mining properties.

Methods of valuation

The process of determining the value or worth of a mining property

- ▶ Income/cash flow approach
- ▶ Market related approach
- ▶ Cost approach
- ▶ Option pricing



Earlier Methods of Mine valuation

- ▶ For mine valuation Hoskold developed the concept of present worth of mining project and written as:

$$Vp = \frac{A}{\frac{i}{(1+i)^n - 1} + i'}$$

1. This formula results in an undervaluation of mineral property
2. Two rates of return

Where, Vp is present worth, i is safe rate on redemption of capital, i' is speculative/risky rate on invested capital and A is uniform annual annuity, n is years of operation

Morkill formula

$$Vp = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$

Income/cash flow approach

- ▶ In the income approach, manager calculate net present value of an investment project and accept the project if it has a positive NPV. In corporate valuation, discounted cash flow model is also called the Net Present Value
- ▶ (NPV) model which is the present value of future cash inflows of an investment or project minus the present and any associated future cash outflow. The NPV is widely used to appraise a project or investment. If the NPV arrives greater than or equal to zero, then theoretically the project or investment should be implemented; if the NPV is less than zero, the project should be discarded
- ▶ The approach assumes that a purchaser would not be justified in paying more to acquire income-producing property than the present value of the income stream to be derived from the property.
- ▶ Because mines have limited operating horizons, and because there are well-established markets for most mineral commodities, the income approach is widely used in valuing mineral properties. The approach is commonly used by the mining industry in the assessment of investment rates of return and to determine appropriate purchase prices for mines or mineral prospects.
- ▶ It depends on reserves, production rate, operating cost, capital cost, environment and redemption cost, commodity price, discount rate

Discounted cash flow (DCF) model reflects the present value of all estimated future cash flow within predetermined budgetary periods and possible continuing value. These cash flows include: operational expenses, sales income, and dividends. The future cash flow are discounted by the discount factor incorporating adjusted risk factor. DCF can be written as below:

$$DPV = \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \dots + \frac{CF_n}{(1+r)^n}$$

Where:

DPV, discounted present value

CF_i , the expected cash flow in period i

r , constant discount rate

The discounted cash flow (DCF) model has evolved over last 60 years and is essentially used to estimate the company's value in terms of time value. Time value is assumed as a passage of time goes by, there should be a given amount of interest or inflation incurred.

Market (Comparable Sales) Approach

- ▶ This approach is an indicator of market value of an item or any project, including a mineral project. The value of a mine is reflected by balance of supply and demand in the market, particularly for an operating mine.
- ▶ This approach is considered by most appraisers and the courts to provide the best indicator of fair market value, since it reflects the balance of supply and demand in the marketplace.
- ▶ The market approach assumes that a purchaser would not be justified in paying more for a property than it would cost him to acquire an equally desirable substitute property. The concept of market value also presumes conditions of an open market, exposure for a reasonable time, knowledgeable buyers and sellers, absence of pressure on either the seller to sell or the buyer to buy, and a sufficient number of transactions to create a stable market.
- ▶ The market approach encounters serious practical problems when applied to mining transactions. This is mainly due to two facts: first, there are very few sales of mining properties, and therefore few comparative data are available; and, second, since each mineral deposit is unique in quality, size, geographical location, degree of development, and many other parameters, any market data are of modest value at best. To be applicable, the market data must not only relate to similar assets but must also be for a similar point in time.
- ▶ Experience in the area of mineral property transactions suggests that the open-market, unpressurized dealing and other assumptions previously mentioned in association with this approach are seldom reflected in reality. When such criteria and assumptions are met, it is often extremely difficult to ascertain the actual or true value of the sale because of stipulations pertaining to production commitments, deferred payments, exchanges of stock, production payments, and other subtle factors that can affect the value significantly.

Cost Approach

- ▶ The cost approach to mine valuation attempts to determine the depreciated replacement cost for the asset in question. That is, what would it cost to reproduce an asset of identical quality and state of repair? The fundamental concept with this approach is that a purchaser would not be justified in paying more for a property than it would cost him to acquire land and construct improvements that had comparable utility with no undue delay.
- ▶ The cost approach is rarely applicable in mining because the correlation between construction costs and the value of the property is very imperfect. For example, if one were to build mines with production capacities of 100 tpd (90 t/day) each, one on a very rich ore deposit and one on an economically marginal deposit, construction costs might be very similar, but fair market values of the two mines would, clearly, be substantially different.
- ▶ Another problem arises when the cost approach is applied to newly discovered mineral properties that have no surface improvements or equipment of any kind. The very nature of mineral exploration and mining dictates that the discovery value of an ore deposit is generally greater than the cost incurred in making that discovery. If this were not true in the aggregate, investment could not be justified for exploration. Furthermore, the notion of estimating the cost of acquiring a comparable asset (ore body) is not very useful. This cost could, for example, be infinite if nature failed to provide a duplicate for explorationists to find.
- ▶ The cost approach is not only the least applicable method in the valuation of mining properties, but it generally is the least reliable also.

Option pricing

- ▶ The standard valuation technique - discounted cash flows (DCF) - estimates value as the present value of expected future cash flows. This technique is typically implemented under an assumption that investment policy is independent of prices. In reality, management responds to fluctuating commodity prices by altering investment policy, such that production expands in the high price scenario and reduces in response to low prices.

The theoretical models of the option pricing model have greatly developed since Merton and Scholes received the Nobel Prize in Economics in 1997. The Black-Scholes model is the most prominent option pricing model and built upon the existing work of Black and Merton model

Approach	Definition	Method
Income based valuation	Determine the value of a business based on its ability to generate desired economic benefit	Discounted Cash flow Method
		Real option valuation
		Earnings multiple
Market based valuation	Establish the business value in comparison to other similar businesses	Comparable method
Cost based valuation	Determine the business value based on the fair market value of its assets less its liabilities	Appraised value

- Merton (1973) showed the value of a call option is an increasing function of rate of interest and time to maturity. **The Black - Scholes model** is a mathematical model used as a financial pricing instrument (Black & Scholes 1973). They show a positive function between the variance of the risky asset and the price of the option and is described in the following formula:

$$C(S, t) = N(d_1) S - N(d_2) K e^{-r(T-t)}$$
$$d_1 = \frac{\ln(\frac{S}{K}) + (r + \frac{\sigma^2}{2})(T - t)}{\sigma \sqrt{T - t}}$$
$$d_2 = \frac{\ln(\frac{S}{K}) + (r - \frac{\sigma^2}{2})(T - t)}{\sigma \sqrt{T - t}} = d_1 - \sigma \sqrt{T - t}.$$

Where:

- S, be the price of the underlying asset
- C(S,t) the price of a European call option .
- K, the strike price
- r, risk free interest
- σ , the volatility of the stock's returns;
- T-t, expiration time in years;

- ▶ It is claimed that valuation procedure related to option pricing theory provides an alternative to discounted cash flow analysis. This procedure is advantageous because it does not require explicit cash flow forecasts or risk adjusted discount rates.
- ▶ The development of valuation procedure by option pricing is to recognize the similarity between a mine and a stock option. The buy or sell option in stocks depends on its price. Similarly, a mine may be opened or closed, depending on the price of the ore and its production cost.
- ▶ The similarity to stock option suggests that there is a relationship between price volatility of minerals. Two mines which produce different minerals but possess identical capacities, cost structures and product prices will be valued differently, if the price volatilities of their mineral product differs.

Purpose Of Mine Valuation Studies

- ▶ Acquisition
- ▶ Taxation
- ▶ Financing
- ▶ **Regulatory Requirements**

Time value of money

- ▶ Time value of money means that a sum of money is worth more now than the same sum of money in the future. This is because money can grow only through investing. An investment delayed is an opportunity lost.
- ▶ The formula for computing the time value of money considers the amount of money, its future value, the amount it can earn, and the time frame.
- ▶ Interest is generally defined as money paid for the use of borrowed money. Interest may be likened to a rental charge for using an asset over some specific time period. The rate of interest is the ratio of the interest chargeable at the end of a specific period of time to the money owed, or borrowed, at the beginning of that period. Interest exists to compensate for a number of concerns experienced by lenders; these are related primarily to risk, inflation, transaction costs, opportunity costs, and postponement of pleasures. The level of interest is, like the price of other assets, determined by supply and demand.

The six basic interest equations are developed and described.

1. Single payment compound amount, $(F/P,i,n)$.

$$F = P (1 + i)^n$$

F is a future sum of money, P is a present sum of money, A is a payment in a series of n equal payments, made at the end of each period of interest, i is effective interest rate per period, and n is number of interest periods.

2. Single payment, present worth $(P/F,i,n)$.

$$P = F/(1 + i)^n$$

3. Uniform series, compound amount $(F/A,i,n)$.

$$F = \frac{A[(1 + i)^n - 1]}{i}$$

If payments of \$725 are made at the end of each year for 12 years to an account which pays interest at the rate of 9% per year, what will be the terminal amount?

4. Uniform series, sinking fund, $(A/F, i, n)$.

$$A = iF / [(1 + i)^n - 1]$$

5. Uniform series, present worth $(P/A, i, n)$.

$$P = \frac{A[(1 + i)^n - 1]}{i(1 + i)^n}$$

6. Uniform series, capital recovery, $(A/P, i, n)$

$$A = \frac{iP(1 + i)^n}{[(1 + i)^n - 1]}$$

With interest at 6%, how much must be deposited at the end of each year to yield a final amount of \$2825 in 7 years?

An investment will yield \$610 at the end of each year for 15 years. If interest is 10%, what is the maximum purchase price (i.e., present value) for this investment?

If an investment opportunity is offered now for \$3500, how much must it yield at the end of every year for 6 years to justify the investment if interest is 12%?

NOMINAL AND EFFECTIVE INTEREST RATES

Interest is generally compounded at the end of each year. In most of the practical evaluation purposes, this provides a satisfactory basis for time value calculation, and is consistent with the end of year convention adoption for cash flows.

However, special situations may arise where there is a need to evaluate a project in more detail by using month by month or quarter by quarter or six monthly distributions by cash flows. When this occurs, there is an important difference between nominal and effective annual rates. This effective rate can be determined as follows:

Let

i = Effective annual interest rate

r = Nominal annual rate

m = Number of compounding periods per year

Then

$$i = (1 + r/m)^n - 1$$

The effective rate is always greater than or equal to the nominal rate and increases as a function of number of compounding periods per year.

Thus, if one charges interest at a compound rate of 2% per month, the nominal rate will be 24%. But the effective rate will be greater than 24% due to compounding effect that occurs every month as determined below:

$$\begin{aligned} i &= (1 + 24/12)^{12} - 1 \\ &= 1.268 - 1 = 26.8\% \end{aligned}$$

Table 5.1 gives the relationship between nominal and effective rates with different compounding frequencies per year.

Table 5.1 Relationship between Nominal and Effective Rates

Compounding frequency	Number of periods per year	Nominal annual rate (%)	Effective annual rate (%)
Annual	1	12	12
Six monthly	2	12	12.36
Quarterly	4	12	12.55
Monthly	12	12	12.68

Illustration 5.6: If an investment opportunity is offered now for ₹5000 how much must it yield at the end of every year for 5 years to justify the investment if interest rate is 10%?

$$A = 5000 \frac{0.1 (1.1)^5}{(1.1)^5 - 1}$$

$$= 1,318.99$$

INFLATION

Inflation deals with the decreased purchasing power of money. It is indicated by the progressive increases in the amount that must be paid for running a business. Figure 5.1 illustrates the effect of inflation expressed as an average rate over a period of time. Let A_1 be the amount of money required after n years for purchasing the same amount of goods that could be purchased by the amount A_0 at the reference time (years) where the increase is expressed at an average rate f . One may write:

$$A_1 = A_0(1 + f)^n$$

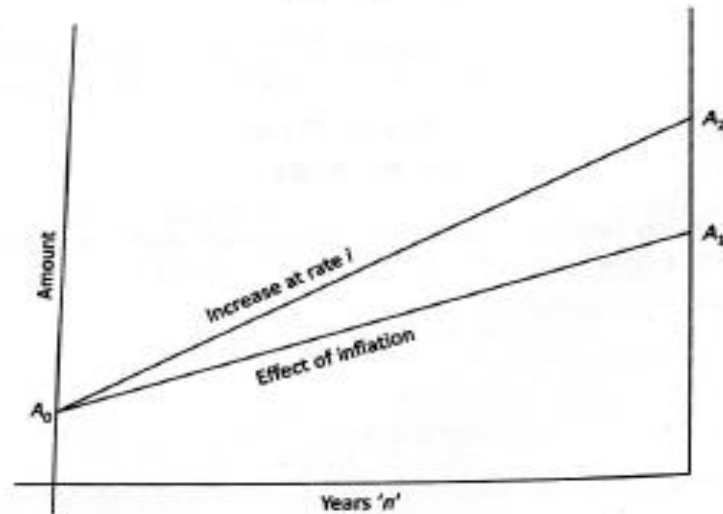


FIGURE 5.1 Effect of inflation.

When the increase from A_0 to A_2 is expressed at an average interest rate i over n years, we have

$$A_2 = A_0 (1 + i)^n$$

The real gain ($s\%$) is only from A_1 to A_2 and if this occurs at a constant rate over n years, we have

$$A_2 = A_1(1 + s)^n$$

or

$$(1 + s) = \frac{(1 + i)}{(1 + f)}$$

or

$$s = \frac{(i - f)}{(1 + f)}$$

If one obtains 10% interest rate and inflation proceeds at the rate of 6%, the actual gain is as follows:

$$s = \frac{0.10 - 0.06}{1 + 0.06} = 0.038 \quad \text{or} \quad 3.8\%$$

High rates of inflation can distort the results of a discounted cash flow, usually making it more attractive than it really is. To explain the inflation effect in cash flow, it can be expressed either in real terms or in money terms.

1. Real term: A cash flow is expressed in real terms when the currency units (rupee, dollar, etc.) have the same buying power throughout the life of the project.

2. Money term: A cash flow is expressed in money terms when all the figures in the cash flow are the actual amount paid or received at the time of the transaction.

Let a five-years project cost ₹3,000 in initial investment and create an income of ₹1,000 per year from the end of year one. The general rate of inflation is 10% per year, i.e., the profits also increase by 10% per year. The required rate of return is 15%.

Some literatures use the terms constant and current money values in place of real term and money term.

Table 5.2 shows the calculation of net present value in real terms.

Table 5.2 NPV Calculation in Real Terms

End of year	Cash flow (₹)	Discount factor at 15%	NPV (₹)
0	3000	1.00	3000
1	1000	0.870	870
2	1000	0.756	756
3	1000	0.658	658
4	1000	0.572	572
5	1000	0.497	497

Present value in real terms = 353.00.

Table 5.3 shows the calculation of net present value in money terms.

2.2 ECONOMIC CONCEPTS INCLUDING CASH FLOW

In Chapter 6, the production planning portion of this text, an economic basis will be used to select production rate, mine life, etc. This section has been included to support that chapter. It is not intended to be a textbook complete in itself but rather to demonstrate some of the important concepts and terms.

2.2.1 Future worth

If someone puts \$1 in a savings account today at a bank paying 10% simple interest, at the end of year 1 the depositor would have \$1.10 in his account. This can be written as

$$FW = PV(1 + i) \quad (2.1)$$

where FW is the future worth, PV is the present value, i is the interest rate.

If the money is left in the account, the entire amount (principal plus interest) would draw interest. At the end of year 2, the account would contain \$1.21. This is calculated using

$$FW = PV(1 + i)(1 + i)$$

At the end of year n , the accumulated amount would be

$$FW = PV(1 + i)^n \quad (2.2)$$

In this case if $n = 5$ years, then

$$FW = \$1(1 + 0.10)^5 = \$1.61$$

2.2.2 Present value

The future worth calculation procedure can now be reversed by asking the question 'What is the present value of \$1.61 deposited in the bank 5 years hence assuming an interest rate of 10%?' The formula is rewritten in the form

$$PV = \frac{FW}{(1 + i)^n} \quad (2.3)$$

Substituting $FW = \$1.61$, $i = 0.10$, and $n = 5$ one finds as expected that the present value is

$$PV = \frac{\$1.61}{(1 + 0.10)^5} = \$1$$

2.2.3 Present value of a series of uniform contributions

Assume that \$1 is to be deposited in the bank at the end of 5 consecutive years. Assuming an interest rate of 10%, one can calculate the present value of each of these payments. These individual present values can then be summed to get the total.

Year 1: Payment

$$PV_1 = \frac{\$1}{(1.10)^1} = \$0.909$$

Year 2: Payment

$$PV_2 = \frac{\$1}{(1.10)^2} = \$0.826$$

Year 3: Payment

$$PV_3 = \frac{\$1}{(1.10)^3} = \$0.751$$

Year 4: Payment

$$PV_4 = \frac{\$1}{(1.10)^4} = \$0.683$$

Year 5: Payment

$$PV_5 = \frac{\$1}{(1.10)^5} = \$0.621$$

The present value of these 5 payments is

$$PV = \$3.790$$

The general formula for calculating the present value of such equal yearly payments is

$$PV = FW \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right] \quad (2.4)$$

Applying the formula in this case yields

$$PV = \$1 \left[\frac{(1.10)^5 - 1}{(0.10)(1.10)^5} \right] = \$3.791$$

The difference in the results is due to roundoff.

2.2.4 Payback period

Assume that \$5 is borrowed from the bank today (time = 0) to purchase a piece of equipment and that a 10% interest rate applies. It is intended to repay the loan in equal yearly payments of \$1. The question is 'How long will it take to repay the loan?' This is called the payback period. The present value of the loan is

$$PV(\text{loan}) = -\$5$$

The present value of the payments is

$$PV(\text{payments}) = \$1 \left[\frac{(1.10)^n - 1}{0.10(1.10)^n} \right]$$

The loan has been repaid when the net present value

$$\text{Net present value (NPV)} = PV(\text{loan}) + PV(\text{payments})$$

is equal to zero. In this case, one substitutes different values of n into the formula

$$NPV = -\$5 + \$1 \left[\frac{(1.10)^n - 1}{0.10(1.10)^n} \right]$$

For $n = 5$ years $NPV = -\$1.209$; for $n = 6$ years $NPV = -\$0.645$; for $n = 7$ years $NPV = -\$0.132$; for $n = 8$ years $NPV = \$0.335$.

Thus the payback period would be slightly more than 7 years ($n \cong 7.25$ years).

2.2.5 Rate of return on an investment

Assume that \$1 is invested in a piece of equipment at time = 0. After tax profits of \$1 will be generated through its use for each of the next 10 years. If the \$5 had been placed in a bank at an interest rate of i then its value at the end of 10 years would have been using Equation (2.2).

$$FW = PV(1 + i)^n = \$5(1 + i)^{10}$$

The future worth (at the end of 10 years) of the yearly \$1 after tax profits is

$$FW = A_m \left[\frac{(1 + i)^n - 1}{i} \right]$$

where A_m is the annual amount and $[(1 + i)^n - 1]/i$ is the uniform series compound amount factor.

The interest rate i which makes the future worths equal is called the rate of return (ROR) on the investment.

In this case

$$\$5(1 + i)^{10} = \$1 \left[\frac{(1 + i)^{10} - 1}{i} \right]$$

Solving for i one finds that

$$i \cong 0.15$$

The rate of return is therefore 15%. One can similarly find the interest rate which makes the net present value of the payments and the investment equal to zero at time $t = 0$.

$$NPV = -\$5 + \$1 \left[\frac{(1.10)^{10} - 1}{i(1 + i)^{10}} \right] = 0$$

$$i \cong 0.15$$

The answer is the same.

The process of bringing the future payments back to time zero is called 'discounting'.

Cash flow (CF)

This is the flow of cash from the project owners for the capital expenditure to get the project into operation (negative) and the flow of cash from the project to the owners after all costs (which should be positive).

The term 'cash flow' refers to the net inflow or outflow of money that occurs during a specific time period. The representation using the word equation written vertically for an elementary cash flow calculation is

$$\begin{array}{rcl} & \text{Gross revenue} & \\ - & \text{Operating expense} & \\ = & \text{Gross profit (taxable income)} & \\ - & \text{Tax} & \\ = & \text{Net profit} & \\ - & \text{Capital costs} & \\ \hline = & \text{Cash flow} & \end{array}$$

A simple example (after Stermole & Stermole, 1987) is given in Table 2.1.

In this case there is a capital expense of \$200 incurred at time $t = 0$ and another \$100 at the end of the first year. There are positive cash flows for years 2 through 6.

Table 2.1. Simple cash flow example (Stermole & Stermole, 1987).

Year	0	1	2	3	4	5	6
Revenue			170	200	230	260	290
– Operating cost			–40	–50	–60	–70	–80
– Capital costs	–200	–100					
– Tax costs			–30	–40	–50	–60	–70
Project cash flow	–200	–100	+100	+110	+120	+130	+140

List of Sequences for
Economic Evaluation

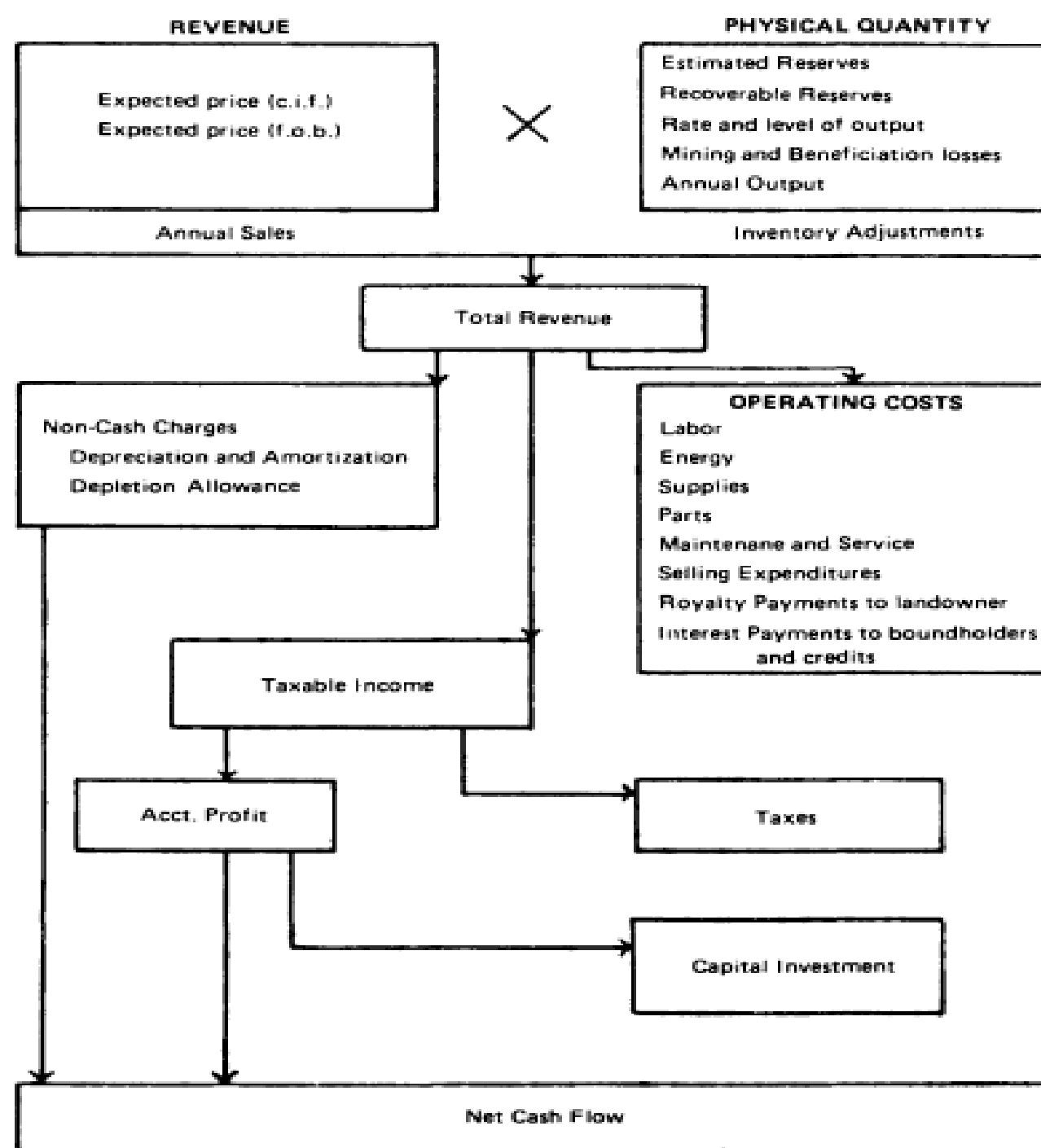


Table 10 : Gross Income, Coal Strip Mine

(Values in \$1,000)

	(1) Gross Revenue		\$37,200
less (1) x .2 =	(2) Royalty Payments		<u>7,440</u>
(1) - (2) =	(3) Working Interest		29,760
	(4) Expensed investment ⁷	0	
	(5) Operating costs	<u>16,770</u>	
(4) + (5) =	(6) Total costs		<u>16,770</u>
(3) - (6) =	(7) Gross earnings		<u>12,990</u>
less	(8) Depreciation allowance		<u>3,358</u>
(7) - (8) =	(9) Balance I		9,632
(3) x .1 =	(10) Depletion allowance		<u>2,976</u>
(9) - (10) =	(11) Taxable income		6,656
(11) x .5 =	(12) Tax liability	3,328	
	(13) Investment tax credit ⁸	<u>24</u>	
(12) - (13) =	(14) Adjusted tax liability		<u>3,304</u>
(11) - (14) =	(15) Net Profit after tax		\$ 3,352

Table 11 : Adjustments for Year 3 from Accounting Profits to Net Cash Flow

(Values in \$1,000)

	(15) Net profit after tax		\$3,352
add	(8) Depreciation allowance	3,358	
add	(10) Depletion allowance	<u>2,976</u>	
(15) + (8) + (10) =	(16) Balance II		<u>9,686</u>
less	(17) Capitalized investment		744
add	(18) Other cash adjustments*		<u>3</u>
(16) - (17) + (18) =	(19) Net cash flow		\$8,945

*Includes either positive or negative cash flows, such as income from salvage, changes in working capital, etc.

Thanks